

SQL assignment

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1. Determine the Structure of DEPT Table and its Contents.

```
SQL> DESC DEPARTMENTS;
Name
-----
DEPARTMENT_ID      NOT NULL NUMBER(4)
DEPARTMENT_NAME     NOT NULL VARCHAR2(30)
MANAGER_ID          NUMBER(6)
LOCATION_ID           NUMBER(4)
```

3. Display the Ename and Deptno from Emp table whose Empno is 7788.

```
SELECT ENAME,DEPTNO FROM EMP WHERE EMPNO=7788;
```

4. Display the Ename,Sal,Comm from Emp table who earn commission and sort the records in descending order of Salary and Comm. Use column's numeric position in the ORDER BY clause

```
SQL> SELECT FIRST_NAME, LAST_NAME, SALARY, COMMISSION_PCT FROM employees WHERE COMMISSION_PCT IS NOT NULL ORDER BY 3 DESC, 4 DESC;
```

FIRST_NAME	LAST_NAME	SALARY	COMMISSION_PCT
John	Singh	10000	.4
Neena	Barrows	12000	.3
Alberto	Erasmus	12000	.3
Lex	De Haan	11000	.25
Gerald	Cash	11000	.3
Ellen	Quart	11000	.3
Clara	Vishney	10500	.25
Eleni	Zlotkey	10500	.2
Samuel	King	10000	.35
Sean	Turner	10000	.3
Marlon	Green	10000	.3
FIRST_NAME	LAST_NAME	SALARY	COMMISSION_PCT
Taylor	Fax	9600	.2
Patricia	Sully	9500	.35
David	Bernstein	9500	.25
Isabelle	Greene	9500	.15
Allan	McEwen	9000	.35
Walter	Whalen	9000	.25
Alyssa	Hutton	8800	.25
Jonathan	Taylor	8800	.2
Jack	Livingston	8600	.3
Lindsey	Smith	8600	.3
Christopher	Olson	8600	.2
FIRST_NAME	LAST_NAME	SALARY	COMMISSION_PCT
Ismael	Green	7800	.3
Neena	Cash	7800	.2
William	Smith	7800	.15
Elizabeth	Bates	7200	.15
Matthew	Marvin	7200	.1
Samuel	Green	7000	.25
Oliver	Tavault	7000	.15
Michael	Grant	7000	.15

6. The HR department wants more descriptive column headings for its report on employees. Name the column headings Emp #, Employee, Job, and Hire Date, respectively by giving Column Aliases.

```
SQL> select EMPLOYEE_ID as EMP#, FIRST_NAME || ' ' || LAST_NAME as EMPLOYEE, JOB_ID as JOB, HIRE_DATE FROM employees;
```

EMP#	EMPLOYEE	JOB	HIRE_DATE
100	Steven King	AD_PRES	17-JUN-13
101	Neena Yang	AD_VP	21-SEP-13
102	Lex Garcia	AD_VP	13-JAN-11
103	Alexander James	IT_PROG	03-JAN-16
104	Bruce Miller	IT_PROG	21-MAY-17
105	David Williams	IT_PROG	23-JUN-15
106	Valli Jackson	IT_PROG	05-FEB-16
107	Olivia Nguyen	IT_PROG	07-FEB-17
108	Nancy Greenberg	FI_MGR	17-AUG-12
109	Daniel Fawcett	FI_ACCOUNT	16-AUG-12
110	John Chen	FI_ACCOUNT	28-SEP-15
EMP#	EMPLOYEE	JOB	HIRE_DATE
111	Ismael Sierra	FI_ACCOUNT	30-SEP-15
112	Jose Manuel Gran	FI_ACCOUNT	07-MAR-16
113	Luis Popp	FI_ACCOUNT	07-DEC-17
114	Den Li	PU_MAN	07-DEC-12
115	Alexander Khoo	PU_CLERK	18-MAY-13
116	Shelli Baida	PU_CLERK	24-DEC-15
117	Sigal Tobias	PU_CLERK	24-JUL-15
118	Guy Himuro	PU_CLERK	15-NOV-16
119	Haven Colmenares	PU_CLERK	20-AUG-17
120	Matthew Weiss	ST_MAN	18-JUL-14

8. To familiarize yourself with the data in the EMP table, create a query to display all the data from that table. Separate each column output by a comma. ENAME, JOB, HIREDATE, MGR. Name the column title THE_OUTPUT.

```
SQL> select FIRST_NAME || ' ' || JOB_ID || ' ' || HIRE_DATE || ' ' || MANAGER_ID as THE_OUTPUT FROM employees;
```

THE_OUTPUT
Steven , AD_PRES , 17-JUN-13 , 100
Neena , AD_VP , 21-SEP-13 , 100
Lex , AD_VP , 13-JAN-11 , 100
Alexander , IT_PROG , 03-JAN-16 , 103
Bruce , IT_PROG , 21-MAY-17 , 105
David , IT_PROG , 23-JUN-15 , 105
Valli , IT_PROG , 05-FEB-16 , 106
Olivia , IT_PROG , 07-FEB-17 , 106
Nancy , FI_MGR , 17-AUG-12 , 108
Daniel , FI_ACCOUNT , 16-AUG-12 , 108
John , FI_ACCOUNT , 28-SEP-15 , 108
THE_OUTPUT
Ismael , FI_ACCOUNT , 30-SEP-15 , 108
Jose Manuel , FI_ACCOUNT , 07-MAR-16 , 108
Luis , FI_ACCOUNT , 07-DEC-17 , 108
Den , PU_MAN , 07-DEC-12 , 108
Alexander , PU_CLERK , 18-MAY-13 , 114
Shelli , PU_CLERK , 24-DEC-15 , 114
Sigal , PU_CLERK , 24-JUL-15 , 114
Guy , PU_CLERK , 15-NOV-16 , 114
Haven , PU_CLERK , 20-AUG-17 , 114

10. Display the ename and department number of all employees in departments 20 or 30 in ascending alphabetical order by ename

```
SQL> select FIRST_NAME, DEPARTMENT_ID FROM employees WHERE DEPARTMENT_ID=20 OR DEPARTMENT_ID=30 ORDER BY FIRST_NAME ASC;
```

FIRST_NAME	DEPARTMENT_ID
Alexander	30
Den	30
Guy	30
Neena	30
Michael	20
Pat	30
Shelli	30
Sigal	30

8 rows selected

12. The HR department needs a report that displays the last name and hire date for all employees who were hired in 1981

```
SQL> SELECT LAST_NAME, HIRE_DATE FROM employees WHERE (HIRE_DATE BETWEEN '01-JAN-1981' AND '31-DEC-1981');
```

no rows selected

2. Determine the Structure of EMP Table and its Contents.

```
SQL> DESC employees;
Name
-----
EMPLOYEE_ID      NOT NULL NUMBER(6)
FIRST_NAME       VARCHAR2(20)
LAST_NAME        NOT NULL VARCHAR2(25)
EMAIL            NOT NULL VARCHAR2(25)
PHONE_NUMBER     VARCHAR2(20)
HIRE_DATE        NOT NULL DATE
JOB_ID           NOT NULL VARCHAR2(10)
SALARY           NUMBER(8, 2)
COMMISSION_PCT   NUMBER(2, 2)
MANAGER_ID       NUMBER(6)
DEPARTMENT_ID    NUMBER(4)
```

SQL> |

5. The HR department needs a query to display all unique job codes from the EMP table

```
SQL> SELECT DISTINCT JOB_ID FROM employees;
```

JOB_ID
AC_ACCOUNT
AC_MGR
AD_ASST
AD_PRES
AD_VP
FI_ACCOUNT
FI_MGR
HR_REP
IT_PROG
MK_MAN
MK_REP
JOB_ID
PR_REP
PU_CLERK
PU_MAN
SA_MAN
SA_REP
SH_CLERK
ST_CLERK
ST_MAN

7. The HR department has requested a report of all employees and their job IDs. Display the last name concatenated with the job ID (separated by a comma and space) and name the column Employee and Title by giving Column Aliases.

```
SQL> select LAST_NAME || ' ' || JOB_ID as EMPLOYEE_AND_TITLE FROM employees;
```

EMPLOYEE_AND_TITLE
Abel , SA_REP
Ande , SA_REP
Atkinson , ST_CLERK
Baida , PU_CLERK
Banda , SA_REP
Bates , SA_REP
Bell , SH_CLERK
Bernstein , SA_REP
Bissot , ST_CLERK
Bloom , SA_REP
Brown , PR_REP
EMPLOYEE_AND_TITLE
Bull , SH_CLERK
Cabrio , SH_CLERK
Cambraut , SA_MAN
Cambraut , SA_REP
Chen , FI_ACCOUNT
Chung , SH_CLERK
Colmenares , PU_CLERK
Davies , ST_CLERK
Davis , MK_REP
Dellinger , SH_CLERK
Dilly , SH_CLERK

9. Create a report to display the ename, job, and Hiredate for the employees name is SCOTT or TURNER. Order the query in ascending order by hiredate

```
SQL> select FIRST_NAME, JOB_ID, HIRE_DATE FROM employees WHERE FIRST_NAME='SCOTT' OR FIRST_NAME='TURNER' ORDER BY HIRE_DATE ASC;
```

no rows selected

11. Modify the previous query to display the last name and salary of employees who earn between 2000 and 3000 and are in department 20 or 30. Label the columns Employee and Monthly Salary, respectively giving Column Aliases.

```
SQL> select LAST_NAME as EMPLOYEE, SALARY as MONTHLY_SALARY, HIRE_DATE, DEPARTMENT_ID FROM employees WHERE (SALARY BETWEEN 2000 AND 3000) AND (DEPARTMENT_ID=20 OR DEPARTMENT_ID=30) ORDER BY HIRE_DATE ASC;
```

EMPLOYEE	MONTHLY_SALARY	HIRE_DATE	DEPARTMENT_ID
Neena	2000	28-JUL-15	30
Baida	2000	24-DEC-15	30
Himuro	2000	15-NOV-16	30
Colmenares	2500	18-AUG-17	30

13. Display the Ename, Sal of employees who earn more than an amount the user specifies after a Prompt.

```
SQL> SELECT FIRST_NAME, SALARY FROM employees WHERE SALARY > &ENTER_SALARY;
Enter value for enter_salary: 5000
old 1: SELECT FIRST_NAME, SALARY FROM employees WHERE SALARY > &ENTER_SALARY
new 1: SELECT FIRST_NAME, SALARY FROM employees WHERE SALARY > 5000
```

FIRST_NAME	SALARY
Steven	24000
Neena	17000
Lex	17000
Alexander	9000
Bruce	6000
Nancy	12000
Daniel	9000
John	8200
Ismael	7700
Jose Manuel	7800
Luis	6900

Alexander	30
Ben	30
Guy	30
Harun	30
Michael	20
Pat	20
Shelli	30
Sigal	30

12.The HR department needs a report that displays the last name and hire date for all employees who were hired in 1981

SQL> SELECT LAST_NAME,HIRE_DATE FROM employees WHERE (HIRE_DATE BETWEEN '01-JAN-1981' AND '31-DEC-1981') ;
no rows selected

SQL> SELECT FIRST_NAME,SALARY FROM employees WHERE SALARY>&ENTER_SALARY;
Enter value for enter_salary: 5000
old 1: SELECT FIRST_NAME,SALARY FROM employees WHERE SALARY>&ENTER_SALARY
new 1: SELECT FIRST_NAME,SALARY FROM employees WHERE SALARY>5000

FIRST_NAME	SALARY
Steven	24000
Neena	17000
Lex	17000
Alexander	9000
Bruce	6000
Nancy	12008
Daniel	9000
John	8200
Ismael	7700
Jose Manuel	7800
Luis	6900

FIRST_NAME	SALARY
Den	11000
Matthew	8000
Adam	8200
Payam	7900
Shanta	6500
Kevin	5800